

Distribution Matching Distillation equation

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1 Introduction

This report is concerned with re-verifying a step in the mathematics of the paper of Yin et al [1]. We wish to verify the following equation (Equations 1-2 in the original paper):

$$D_{\text{KL}}(p_{\text{fake}}||p_{\text{real}}) = \mathbb{E}_{x \sim p_{\text{fake}}} \left[\log \left(\frac{p_{\text{fake}}(x)}{p_{\text{real}}(x)} \right) \right] \quad (1)$$

$$\implies \nabla_{\theta} D_{\text{KL}}(p_{\text{fake}}||p_{\text{real}}) = \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[-(\nabla_x \log p_{\text{real}}(G_{\theta}(z)) - \nabla_x \log p_{\text{fake}}(G_{\theta}(z))) \frac{dG}{d\theta} \right]. \quad (2)$$

G_{θ} is a generative model parametrised by θ , p_{fake} is the probability density function of the images produced by G_{θ} and p_{real} is the probability density function of the distribution of real images in a database.

We have $x = G_{\theta}(z)$ by definition, with $z \sim \mathcal{N}(0, I_d)$. For brevity's sake, let us use the following abbreviations:

- $p_r := p_{\text{real}}$
- $p_f := p_{\text{fake}}$

We consider that $G_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}^d$.

2 Proof

First, of all, we develop:

$$D_{\text{KL}}(p_f||p_r) = \mathbb{E}_{x \sim p_f} \left[\log \left(\frac{p_f(x)}{p_r(x)} \right) \right] \quad (3)$$

$$= \mathbb{E}_{x \sim p_f} [\log p_f(x) - \log p_r(x)] \quad (4)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\log p_f(G_{\theta}(z)) - \log p_r(G_{\theta}(z))] \quad (5)$$

Thus, we have, assuming sufficient regularity of p_f^θ , p_r and G_θ in order to interchange the gradient and expectation, we have:

$$\nabla_\theta D_{\text{KL}}(p_f \| p_r) = \nabla_\theta \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\log p_f(G_\theta(z)) - \log p_r(G_\theta(z))] \quad (6)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\nabla_\theta \log p_f(G_\theta(z)) - \nabla_\theta \log p_r(G_\theta(z))] \quad (7)$$

Therefore, the problem boils down to calculating the following quantities:

- $\nabla_\theta \log p_f(G_\theta(z))$
- $\nabla_\theta \log p_r(G_\theta(z))$

There is a subtlety in the first term: indeed p_f is the distribution of $G_\theta(z)$ with $z \sim \mathcal{N}(0, I_d)$. Thus, the first quantity depends on θ both because of the argument of p_f and because of p_f itself. We will deal with this term second, and treat the easier case first.

2.1 Calculation of $\nabla_\theta \log p_r(G_\theta(z))$

We have:

$$\nabla_\theta \log p_r(G_\theta(z)) = \frac{1}{p_r(G_\theta(z))} \nabla_\theta p_r(G_\theta(z)) \quad (8)$$

We can consider that $p_r(G_\theta(z))$ is the concatenation of two functions: p_r and G_θ . Thus, the chain rule gives:

$$\nabla_\theta \log p_r(G_\theta(z)) = \frac{1}{p_r(G_\theta(z))} \frac{dp_r(x)}{dx} \Big|_{x=G_\theta(z)} \frac{dG_\theta(z)}{d\theta} \quad (9)$$

$$= \frac{1}{p_r(G_\theta(z))} \nabla_x p_r(G_\theta(z)) \frac{dG_\theta(z)}{d\theta}, \quad (10)$$

where $\nabla_x p_r(G_\theta(z))$ should be read as the gradient of the real-valued function $p_r : \mathbb{R}^d \rightarrow \mathbb{R}$, evaluated at the point $x = G_\theta(z)$.

Now, let us note that:

$$\nabla_x \log p_r(G_\theta(z)) = \frac{1}{p_r(G_\theta(z))} \nabla_x p_r(G_\theta(z)). \quad (11)$$

Thus, we have:

$$\nabla_\theta \log p_r(G_\theta(z)) = \nabla_x \log p_r(G_\theta(z)) \frac{dG_\theta(z)}{d\theta}. \quad (12)$$

2.2 Calculation of $\nabla_\theta \log p_f(G_\theta(z))$

Now, let us turn to the more subtle case. We have

$$\nabla_\theta \log p_f(G_\theta(z)) = \frac{1}{p_f(G_\theta(z))} \frac{dp_f(x)}{dx} \Big|_{x=G_\theta(z)} \frac{dG_\theta(z)}{d\theta} \quad (13)$$

$$= \frac{1}{p_r(G_\theta(z))} \nabla_\theta p_f(G_\theta(z)) \quad (14)$$

Now, the difficult part is determining $\nabla_\theta p_f(G_\theta(z))$. $p_f(G_\theta(z))$ can be viewed as a function of two variables:

$$g(\theta, G_\theta(z)) := p_f(G_\theta(z)) \quad (15)$$

$$= G_\theta \# \mathcal{N}(0, I_d), \quad (16)$$

where $G_\theta \# \mathcal{N}(0, I_d)$ is the pushforward measure of the function G_θ applied to the distribution $\mathcal{N}(0, I_d)$. Let us also write $g(x_1, x_2)$ to distinguish between the first and second variables. Finally, when referring to p_f , we will now write p_f^θ , to make the dependence on θ clear.

Using the chain rule for functions of several variables, we have:

$$\nabla_\theta g(\theta, G_\theta(z)) = \nabla_\theta g(\theta, G_\theta(z)) = \frac{dg(\theta, G_\theta(z))}{dx_1} \frac{dx_1|_{x_1=\theta}}{d\theta} + \frac{dg(\theta, G_\theta(z))}{dx_2} \frac{dx_2}{d\theta} \quad (17)$$

$$= \frac{dg(\theta, G_\theta(z))}{d\theta} \cdot 1 + \frac{dg(\theta, G_\theta(z))}{dx_2} \frac{dx_2}{d\theta} \quad (18)$$

$$= \nabla_\theta p_f^\theta(x)|_{x=G_\theta(z)} + \frac{dg(x_1 = \theta, G_\theta(z))}{dx_2} \frac{dG_\theta(z)}{d\theta} \quad (19)$$

$$= \nabla_\theta p_f^\theta(x)|_{x=G_\theta(z)} + \nabla_x p_f^\theta(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \quad (20)$$

It is important to distinguish between the two terms of this last equation. $\nabla_\theta p_f^\theta(x)|_{x=G_\theta(z)}$ represents the differential of p_f wrt θ , where θ is variable for the definition of p_f^θ , but is considered fixed for $x = G_\theta(z)$, thus the notation. Conversely, for the second term, $\nabla_x p_f^\theta(G_\theta(z))$ is the gradient of the distribution p_f^θ , considering θ fixed.

So, coming back to $\nabla_\theta \log p_f(G_\theta(z))$:

$$\nabla_\theta \log p_f(G_\theta(z)) = \frac{1}{p_f(G_\theta(z))} \nabla_\theta p_f(G_\theta(z)) \quad (21)$$

$$= \frac{1}{p_f(G_\theta(z))} \nabla_\theta g(\theta, G_\theta(z)) \quad (22)$$

$$= \frac{1}{p_f(G_\theta(z))} \left(\nabla_\theta p_f^\theta(x)|_{x=G_\theta(z)} + \nabla_x p_f^\theta(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \right) \quad (23)$$

Now let us get back to the full derivation, and note $\nabla_\theta p_f^\theta(x)|_{x=G_\theta(z)} = \nabla_\theta p_f^\theta(G_\theta(z))$ for simplicity, trusting that the variable quantity is clear.

2.3 Final derivation

We have:

$$\nabla_\theta D_{\text{KL}}(p_f \| p_r) = \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\nabla_\theta \log p_f(G_\theta(z)) - \nabla_\theta \log p_r(G_\theta(z))] \quad (24)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[\frac{1}{p_f^\theta(G_\theta(z))} \left(\nabla_\theta p_f^\theta(G_\theta(z)) + \nabla_x p_f^\theta(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \right) - \nabla_x \log p_r(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \right] \quad (25)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[\nabla_\theta \log p_f^\theta(G_\theta(z)) + \nabla_x \log p_f^\theta(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} - \nabla_x \log p_r(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \right] \quad (26)$$

where we have again used $\frac{1}{p_f^\theta(G_\theta(z))} \nabla_\theta p_f^\theta(G_\theta(z)) = \nabla_\theta \log p_f^\theta(G_\theta(z))$, and similarly for ∇_x . But, we also have that:

$$\mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\nabla_\theta p_f^\theta(G_\theta(z))] = \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} [\nabla_\theta \log p_f^\theta(G_\theta(z))] \quad (27)$$

$$= \int p_f^\theta(G_\theta(z)) \nabla_\theta \log p_f^\theta(G_\theta(z)) dz \quad (28)$$

$$= \int \frac{p_f^\theta(x)}{p_f^\theta(x)} \nabla_\theta p_f^\theta(x) dx \quad (29)$$

$$= \nabla_\theta \int p_f^\theta(x) dx \quad (30)$$

$$= \nabla_\theta 1 \quad (31)$$

$$= 0. \quad (32)$$

Thus, we have:

$$\nabla_\theta D_{\text{KL}}(p_f \| p_r) = \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[\nabla_x \log p_f^\theta(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} - \nabla_x \log p_r(G_\theta(z)) \frac{dG_\theta(z)}{d\theta} \right] \quad (33)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[\left(\nabla_x \log p_f^\theta(G_\theta(z)) - \nabla_x \log p_r(G_\theta(z)) \right) \frac{dG_\theta(z)}{d\theta} \right] \quad (34)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[\left(\nabla_x \log p_f^\theta(G_\theta(z)) - \nabla_x \log p_r(G_\theta(z)) \right) \frac{dG_\theta(z)}{d\theta} \right] \quad (35)$$

$$= \mathbb{E}_{z \sim \mathcal{N}(0, I_d)} \left[- \left(\nabla_x \log p_r(G_\theta(z)) - \nabla_x \log p_f^\theta(G_\theta(z)) \right) \frac{dG_\theta(z)}{d\theta} \right], \quad (36)$$

which is the result shown in the paper.

References

- [1] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Fredo Durand, William T Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6613–6623. IEEE, 2024.